ADITYA C. DOPPALAPUDI

Dallas, TX| 757-754-6348 | aditya.doppalapudi@gmail.com | U.S. Citizen | Open to Relocation
https://www.linkedin.com/in/adityadoppalapudi/ | https://github.com/adidoppalapudi

AI/Software Engineer with enterprise experience in conversational AI, AWS deployment, UAT, LLM response validation, and defect resolution. M.S. Computer Science graduate with hands-on experience in Generative AI, RAG/LLM evaluation, AI reliability, Python, cloud AI workflows, and LLM security. Built a Vertex AI and CodeQL secure code-generation pipeline that reduced vulnerability rates by approximately 65% across 1,071 samples.

EDUCATION

Old Dominion University Norfolk, VA
Master of Science in Computer Science (MSCS) – GPA 3.97 Graduated: May 2026
Relevant Coursework: Generative AI; Data Science; Advanced Databases; Big Data; Quantum Computing

Jawaharlal Nehru Technological-UCEST Hyderabad
Bachelor of Technology in Electronics and Communication Engineering (ECE) Graduated: July 2024

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Python, SQL, TypeScript, JavaScript, PostgreSQL, MySQL, MATLAB, C, HTML/CSS, Node.js, RESTful APIs
Tools & Technologies: Claude Code, Codex, GitHub, Git, Linux, Docker, React, Next.js, FastAPI, Tailwind CSS, Google Cloud Vertex AI, AWS, Amazon Bedrock, CodeQL, ArcGIS
AI & Engineering: Generative AI, LLMs, RAG, Software Engineering, Static Analysis, UAT, Functional Testing, Data Structures & Algorithms, SDLC
Certifications & Training: IBM Basics of Quantum Information; IBM Intro to Practical Quantum-Safe Cryptography

WORK EXPERIENCE

OSI Digital **Frisco, TX**
Digital Engineer September 2026 – Present
Project: AI Procurement Assistant | AI Software Engineering, Reliability & UAT

- Working on a conversational AI solution on an enterprise AI procurement platform integrating ERP systems, PostgreSQL, FastAPI, Amazon Bedrock, and AWS services
- Supporting AWS deployment and UAT, including role-based functional testing, ERP data reconciliation, LLM response validation (DeepEval) and defect tracking
- Analyzed system architecture, data workflows and deployment requirements; reviewed 8 AWS CDK stacks covering CloudFormation, VPC networking, IAM policies, and service dependencies for deployment readiness
- Diagnosed and resolved application defects using React, Next.js, Tailwind CSS, FastAPI, PostgreSQL

Old Dominion University **Norfolk, VA**
Graduate Teaching Assistant January 2025 - May 2025

- Facilitated study sessions and provided academic support in Discrete Structures and Computer Architecture, explaining foundational computing, logic, and architecture concepts.
- Evaluated assignments and examinations, providing feedback to reinforce students' technical understanding

Institute of Electrical and Electronics Engineers - Circuits and Systems Society (IEEE-CASS)
VLSI Intern October 2023 - December 2023

- Optimized FPGA-based RSA cryptographic hardware using Kogge-Stone adders and Montgomery multiplication to improve computational efficiency.
- Achieved a **3x reduction in delay** using a 4-bit Kogge Stone adder and approximately **34% efficiency improvement** through Montgomery multiplier implementation

PROJECTS

LLM-Based Secure Coding Assistant | Python, CodeQL, Google Vertex AI, LLMs

- Developed an **LLM-based secure code generation pipeline** using Python and CodeQL, implementing both **Proactive Vulnerability Prevention** and **Post-Hoc Vulnerability Repair** techniques.
- Designed and implemented a **CWE-based vulnerability hint generator** to guide LLMs toward safer code generation.
- Engineered an automated **static analysis feedback loop** using CodeQL findings and LLM-generated explanations to detect, explain, and repair vulnerable code which is integrated into the pipeline.
- Evaluated the pipeline across 1,071 code samples using TarV-R and AllV-R metrics, achieving approximately 65% lower AllV-R vulnerability rate with Google Cloud Vertex AI-based LLM workflows.

Movie Recommendation Engine | Python, FastAPI, PostgreSQL, Pandas, scikit-learn

- Built a web application using the MovieLens 25M dataset, supporting title and genre search, popularity recommendations with year and rating filters, TF-IDF similarity recommendations, analytics dashboards, and movie posters retrieved through the TMDB API.
- Processed 25M ratings into **Parquet** and **PostgreSQL** materialized views, then evaluated Bayesian popularity and rank-50 SVD models on a shared 20% holdout, improving RMSE from 0.9671 to 0.8301 with 0.9248 NDCG@10.

Restoration Efforts in USA| ArcGIS, Environmental Analysis, Python

- Analyzed U.S. restoration project datasets using ArcGIS and Python to identify spatial, temporal, and policy-driven trends across West Coast restoration efforts, including federal and NGO-led initiatives.
- Cleaned, structured, and transformed environmental datasets for geospatial analysis, applying data preprocessing, feature organization, and trend comparison to support project effectiveness and resource allocation insights.

LEADERSHIP & STUDENT ENGAGEMENT

Association for Computing Machinery [ACM] - Student Chapter of ODU Event Coordination Officer from 2024-2026
Global Student Friendship Group Student Leader for Outreach from Oct 2024- May 2026
Institute of Electrical and Electronics Engineers Member Since 2022